

Structure of complete metric space:-

Dense set:-

Let (X, d) be a metric space then $A \subset X$ is called dense in X if $\bar{A} = X$, where $\bar{A} = A \cup A^d$. Equivalently we can say $A \cap U \neq \emptyset$ for any open set U in X .

problem:-

show that $S = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid f(x, y, z) = x^4 + \log(1+y^2) - 5x^3 + 2x^2 - 3z = 0 \right\}$

is not dense in $\mathbb{R}^3$.

[closed set never be a dense set]

$\Rightarrow$ We have to show that S is closed set.

$$S = \{ \bar{x} : f(\bar{x}) = g(\bar{x}) \}$$

$$\text{Here } f(\bar{x}) = x^4 + \log(1+y^2) - 5x^3 + 2x^2 - 3z = 0$$

$$= \{ \bar{x} : (f-g)(\bar{x}) = 0 \}$$

$$g(\bar{x}) = 0.$$

$$= (f-g)^{-1} \{0\}.$$

S is closed as inverse image of the closed set $\{0\}$ under the continuous funcⁿ f is closed.

Nowhere dense:-

Let (X, d) be a metric space and $F \subset X$. F is called nowhere dense set if $(\bar{F})^\circ = \emptyset$ i.e. closure of F contains no interior pts.

Category I & category II set:-

A subset E of X is said to be of category I if it can be written as a countable union of nowhere dense set F_n i.e. $E = \bigcup_{n \geq 1} F_n$ with $(\bar{F}_n)^\circ = \emptyset \forall n = 1, 2, 3, \dots$

A set that is not of category I is said to be a category II set.

Remarks:-

① $F \subset X$ is nowhere dense in X iff $(\overline{F})^c$ is ~~set~~ dense in X i.e. $(\overline{(\overline{F})^c}) = X$

$$\Rightarrow (\overline{X - F}) = X$$

② For an open subset U of X , U is dense in X iff U^c is nowhere dense in X .

③ $F \subset X$ is ~~non~~ nowhere dense iff for every non-empty open set U in X $\exists$ a non-empty open set V in U such that $V \cap F = \emptyset$.

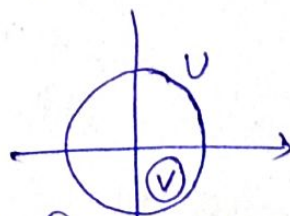
④



Ex-①

Any line ~~in~~ in $\mathbb{R}^2$ is nowhere dense.

Here $V \cap F = \emptyset$ $F = \{y=0\}$



Ex-②

Set of integer and natural no. is nowhere dense.

Ex-③

$$F = \left\{ \frac{1}{n} : n = 1, 2, \dots \right\} \cup \{0\}$$

$$\overline{F} = F \neq \mathbb{R} \Rightarrow (\overline{F})^c = \emptyset$$

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Hence, F is nowhere dense.

Ex-④

In discrete metric space the only nowhere dense is $\emptyset$.

Notes

Not being nowhere dense $\nRightarrow$ everywhere dense.

For example consider $F = \{x \in \mathbb{R} \mid 1 < x < 2\} = (1, 2)$

$$\bar{F} = [1, 2] \Rightarrow \bar{F} \neq \mathbb{R}$$

$\Rightarrow F$ is not dense in $\mathbb{R}$

$$\text{Also } (\bar{F})^\circ = (1, 2) \neq \emptyset$$

$\Rightarrow F$ is not no-where dense in $\mathbb{R}$.

Baire-category theorem :-

Motivation :-

① Let X be a metric space if U & V are open and dense in X then $U \cap V$ is also open & dense in X .

Let G_i be an any open set in X $U \cap G_i \neq \emptyset$ & $V \cap G_i \neq \emptyset$

Now, $(U \cap V) \cap G_i = (U \cap G_i) \cap (V \cap G_i) \neq \emptyset \cap \emptyset = \emptyset$

$\therefore (U \cap V)$ is dense set, by definition.

② On the other hand if U & V are assumed to be dense but not necessary open then $U \cap V$ does not have to be dense in X .

For example $U = \mathbb{Q}$, $V = \mathbb{Q}^c$

U & V are dense in $\mathbb{R}$ but $U \cap V = \emptyset$.

i.e. $U \cap V$ is not dense in $\mathbb{R}$.

③ Consider now the sequence of dense & open set $\{U_n\}_{n \geq 1}$. In general $\bigcap_{n \geq 1} U_n = \emptyset$

Ex :- Consider $(\mathbb{Q}, d)$ with the usual metric d .
Let $\{q_n : n = 1, 2, 3, \dots\}$ be an enumeration of rational number and let $U_n = \mathbb{Q} - \{q_n : n = 1, 2, 3, \dots\}$.
Here each U_n is open ($\because U_n$ is complement of closed set $\{q_n\}$) and dense also.

However $\bigcap_{n \geq 1} U_n = \bigcap_{n \geq 1} (\mathbb{Q} - \{q_n\})$

$$= \mathbb{Q} - \bigcup_{n \geq 1} \{q_n\} = \emptyset$$

The Baire - category theorem says that if (X, d) is complete then $\bigcap_{n \geq 1} U_n$ is complete in X .

Statement:- $\{U_n\}$

Let sequence $\{U_n\}_{n \geq 1}$ be a sequence of open & dense subset in a complete metric space (X, d) then $\bigcap_{n \geq 1} U_n$ is dense in X .

proof:- It is sufficient to show that every open ball $B(x, r)$ contains a point belonging to $U = \bigcap_{n \geq 1} U_n$ i.e. $B(x, r) \cap U \neq \emptyset \quad \forall x \in X$.

Since U_1 is open and dense, $B(x, r) \cap U_1 \neq \emptyset$ also open.

So, there exist an open ball $B(x_1, R)$ with $R < r$ such that $B(x_1, R) \subseteq B(x, r) \cap U_1$.

$$\Rightarrow B(x_1, R) \subseteq B(x, r)$$

$$B(x_1, R) \subseteq U_1$$

Taking $r_1 < R$ we get $\bar{B}(x_1, r_1) \subseteq B(x, r)$

$$\bar{B}(x_1, r_1) \subseteq U_1$$

Similarly since U_2 is open & dense $\exists x_2, r_2 < \frac{1}{2}$ such that $\bar{B}(x_2, r_2) \subseteq \bar{B}(x_1, r_1) \cap U_2$

Continuing in this process we find a sequence of closed ball $\bar{B}(x_n, r_n)$, $r_n < \frac{1}{n}$ and

$$\bar{B}(x_{n+1}, r_{n+1}) \subseteq \bar{B}(x_n, r_n) \cap U_{n+1}$$

We claim that $\{x_n\}$ is Cauchy.

By construction $\forall n \geq k \Rightarrow \bar{B}(x_n, r_n) \subseteq \bar{B}(x_k, r_k)$

Now given $\epsilon > 0$, choose $k = 1, 2, 3, \dots$ st. $\frac{1}{k} < \frac{\epsilon}{2}$

then $n, m \geq k \Rightarrow d(x_n, x_m) \leq d(x_n, x_k) + d(x_k, x_m) < \frac{1}{k} + \frac{1}{k} < \epsilon$

Since X is complete sequence $\{x_n\}$ converge to y (say).
The point $y \in \bar{B}(x_k, r_k)$ since $x_n \in B(x_k, r_k) \forall n \geq k$
and $\bar{B}(x_k, r_k)$ is closed $\forall k$.

After taking a limit as $n \rightarrow \infty$, $y \in \bar{B}(x_k, r_k) \forall k$
in particular $y \in \bar{B}(x_1, r_1)$ which is a subset of $B(x, r)$
and $y \in \bar{B}(x_{n+1}, r_{n+1}) \subset U_{n+1} \forall n \Rightarrow y \in \bigcap_{n=1}^{\infty} U_n = \emptyset \cup$
consequently $y \in B(x, r) \cap \{\bigcap_{n=1}^{\infty} U_n\}$

$$\Rightarrow y \in B(x, r) \cap U \neq \emptyset$$

$$\Rightarrow B(x, r) \cap U \neq \emptyset \quad \forall x \in X \text{ ----- } (*)$$

$$\Rightarrow U \text{ is dense in } X.$$

Corr:- Let (X, d) be a complete metric space and also let
 $E_1, E_2, \dots, E_n$ be a ~~rapid~~ sequence of nowhere dense
set in X . Then $X \neq \bigcup_{n=1}^{\infty} E_n$.

In other words, a complete metric space cannot be covered
by countable number of nowhere dense sets i.e a
complete metric space is of second category.

proof:- We use the following lemma: A set E in X is
nowhere dense iff $(\bar{E})^c$ is dense in X . The sets
 $\{E_n\}$ are nowhere dense meaning that $(\bar{E}_n)^c$ are
dense in X .

Let us define $U_n = (\bar{E}_n)^c \quad \forall n \geq 1$ and open
Since X is complete and the sets U_n are dense in X
Hence $\bigcap U_n$ is dense in X

$$\therefore \bigcap U_n \neq \emptyset, \text{ by } (*)$$

$$\Rightarrow \left(\bigcup_{n=1}^{\infty} U_n^c \right)^c \neq \emptyset^c$$

$$\Rightarrow X \neq \bigcup_{n=1}^{\infty} U_n^c \Rightarrow \bigcup \bar{E}_n \neq X.$$

Since $E_n \subseteq \bar{E}_n \forall n$, $X \neq \bigcup_{n=1}^{\infty} E_n \Rightarrow X$ is of 2nd category.

Remark:-

① Is $\mathbb{R}$ 1st category?

$\Rightarrow$ Since $\mathbb{R}$ is complete, by Baire-category theorem, $\mathbb{R}$ is of ^{2nd} category.

② Is $\mathbb{R}^2$ 1st category?

~~show that
 $\mathbb{R}$ is
nowhere
dense.~~

③ Show that $\mathbb{R}$ is dense.

$$\Rightarrow F = \mathbb{R}$$

$$\bar{F} = \mathbb{R}$$

$\therefore \mathbb{R}$ is dense.

$$\boxed{\bar{F} = \mathbb{R} \neq \emptyset}$$